

ASSIGNMENT 7

E1.

- a) Code (Đã đăng trên github)
- b) Kết quả:
 - Compile:

```
^ Nhung/Assignment 7/E1 >  
gcc -Wall E1.c -o E1.o  
^ Nhung/Assignment 7/E1 >  
|
```

- Chạy code:

```
^ Nhung/Assignment 7/E1 >  
gdb E1.o  
GNU gdb (GDB) 17.1  
Copyright (C) 2025 Free Software Foundation, Inc.  
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>  
This is free software: you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
Type "show copying" and "show warranty" for details.  
This GDB was configured as "x86_64-pc-linux-gnu".  
Type "show configuration" for configuration details.  
For bug reporting instructions, please see:  
<https://www.gnu.org/software/gdb/bugs/>.  
Find the GDB manual and other documentation resources online at:  
  <http://www.gnu.org/software/gdb/documentation/>.  
  
For help, type "help".  
Type "apropos word" to search for commands related to "word" ...  
Reading symbols from E1.o ...  
  
This GDB supports auto-downloading debuginfo from the following URLs:  
  <https://debuginfod.archlinux.org>  
Enable debuginfod for this session? (y or [n]) y  
Debuginfod has been enabled.  
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.  
(No debugging symbols found in E1.o)  
(gdb) r  
Starting program: /home/tranduy/Nhung/Assignment 7/E1/E1.o  
[Thread debugging using libthread_db enabled]  
Using host libthread_db library "/usr/lib/libthread_db.so.1".  
  
--- Current Phonebook ---  
Bucket [0]: (Hoa: 0906) -> NULL  
Bucket [1]: NULL  
Bucket [2]: NULL  
Bucket [3]: NULL  
Bucket [4]: (Nam: 0908) -> NULL  
Bucket [5]: (Linh: 0907) -> (Binh: 0902) -> (An: 0901) -> NULL  
Bucket [6]: (Giang: 0905) -> (Chi: 0903) -> NULL  
Bucket [7]: NULL  
Bucket [8]: (Dung: 0904) -> NULL  
Bucket [9]: NULL  
Found An: 0901  
Found Linh: 0907  
Xuan not found in phonebook.  
Deleted contact: An  
Deleted contact: Dung  
  
--- Current Phonebook ---  
Bucket [0]: (Hoa: 0906) -> NULL  
Bucket [1]: NULL  
Bucket [2]: NULL  
Bucket [3]: NULL  
Bucket [4]: (Nam: 0908) -> NULL  
Bucket [5]: (Linh: 0907) -> (Binh: 0902) -> NULL  
Bucket [6]: (Giang: 0905) -> (Chi: 0903) -> NULL  
Bucket [7]: NULL  
Bucket [8]: NULL  
Bucket [9]: NULL  
[Inferior 1 (process 100873) exited normally]  
(gdb) |
```

E2.

- a) Code: (Đã đăng trong github)

b) Kết quả:

- Compile:

```
^ Assignment 7/E2/E2.2 >  
gcc -c ../E2.1/hashtable.c -o hashtable.o  
^ Assignment 7/E2/E2.2 >  
ar rcs libhash.a hashtable.o  
^ Assignment 7/E2/E2.2 >  
gcc ../E2.1/main.c -L. -lhash -o main.static  
^ Assignment 7/E2/E2.2 >  
|
```

- Kết quả chương trình:

```
^ Assignment 7/E2/E2.2 >  
gdb main.static  
GNU gdb (GDB) 17.1  
Copyright (C) 2025 Free Software Foundation, Inc.  
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>  
This is free software: you are free to change and redistribute it.  
There is NO WARRANTY, to the extent permitted by law.  
Type "show copying" and "show warranty" for details.  
This GDB was configured as "x86_64-pc-linux-gnu".  
Type "show configuration" for configuration details.  
For bug reporting instructions, please see:  
<https://www.gnu.org/software/gdb/bugs/>.  
Find the GDB manual and other documentation resources online at:  
  <http://www.gnu.org/software/gdb/documentation/>.  
  
For help, type "help".  
Type "apropos word" to search for commands related to "word" ...  
Reading symbols from main.static ...  
  
This GDB supports auto-downloading debuginfo from the following URLs:  
  <https://debuginfod.archlinux.org>  
Enable debuginfod for this session? (y or [n])  
Debuginfod has been disabled.  
To make this setting permanent, add 'set debuginfod enabled off' to .gdbinit.  
(No debugging symbols found in main.static)  
(gdb) r  
Starting program: /home/tranduy/Nhung/Assignment 7/E2/E2.2/main.static  
[Thread debugging using libthread_db enabled]  
Using host libthread_db library "/usr/lib/libthread_db.so.1".  
Found: An - 0901  
Contact An not found.  
[Inferior 1 (process 116854) exited normally]  
(gdb) |
```

E3.

a) Code: (Đã đăng trong github)

b) Kết quả:

- Compile:

```
^ Nhung/Assignment 7/E3 >  
gcc -Wall E3.c -o E3.o  
^ Nhung/Assignment 7/E3 >  
|
```

- Kết quả chương trình:

```

A Nhung/Assignment 7/E3 >
gdb E3.o
GNU gdb (GDB) 17.1
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-pc-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
  <http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word" ...
Reading symbols from E3.o ...

This GDB supports auto-downloading debuginfo from the following URLs:
  <https://debuginfod.archlinux.org>
Enable debuginfod for this session? (y or [n]) y
Debuginfod has been enabled.
To make this setting permanent, add 'set debuginfod enabled on' to .gdbinit.
(No debugging symbols found in E3.o)
(gdb) r
Starting program: /home/tranduy/Nhung/Assignment 7/E3/E3.o
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/libthread_db.so.1".
Adding Nguyen Van A ...
Current highest priority: Nguyen Van A (Priority 2)
Adding Tran Thi B ...
Current highest priority: Tran Thi B (Priority 5)
Adding Le Van C ...
Current highest priority: Tran Thi B (Priority 5)
Adding Pham Van D ...
Current highest priority: Tran Thi B (Priority 5)
Adding Hoang Thi E ...
Current highest priority: Tran Thi B (Priority 5)

--- Order of Treatment ---
Calling Patient: Tran Thi B      | Priority: 5
Calling Patient: Pham Van D     | Priority: 4
Calling Patient: Hoang Thi E    | Priority: 3
Calling Patient: Nguyen Van A   | Priority: 2
Calling Patient: Le Van C       | Priority: 1
[Inferior 1 (process 120001) exited normally]
(gdb) █

```